

OS_ORANGE_REPORT

PES2UG24CS042 | Akash Singh

1A

```
akash@LAPTOP-NAL20LH5:/mnt/d/OS_orange/PES2UG24CS042-pes-vcs$ ./test_objects
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
akash@LAPTOP-NAL20LH5:/mnt/d/OS_orange/PES2UG24CS042-pes-vcs$ |
```

1B

```
akash@LAPTOP-NAL20LH5:/mnt/d/OS_orange/PES2UG24CS042-pes-vcs$ find .pes/objects -type f
.pes/objects/2c/f8d83d9ee29543b34a87727421fdec7e3f3a183d337639025de576db9ebb4
.pes/objects/e0/0c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f
akash@LAPTOP-NAL20LH5:/mnt/d/OS_orange/PES2UG24CS042-pes-vcs$ |
```

2A and 2B

```
akash@LAPTOP-NAL20LH5:/mnt/d/OS_orange/PES2UG24CS042-pes-vcs$ make test_tree
gcc -Wall -Wextra -O2 -c test_tree.c -o test_tree.o
gcc -Wall -Wextra -O2 -c tree.c -o tree.o
gcc -o test_tree test_tree.o object.o tree.o -lcrypto
akash@LAPTOP-NAL20LH5:/mnt/d/OS_orange/PES2UG24CS042-pes-vcs$ ./test_tree
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
akash@LAPTOP-NAL20LH5:/mnt/d/OS_orange/PES2UG24CS042-pes-vcs$ make test_objects
```

3A

```
akash@LAPTOP-NAL20LH5:/mnt/d/OS_orange/PES2UG24CS042-pes-vcs$ echo "hello" > file1.txt
akash@LAPTOP-NAL20LH5:/mnt/d/OS_orange/PES2UG24CS042-pes-vcs$ echo "world" > file2.txt
akash@LAPTOP-NAL20LH5:/mnt/d/OS_orange/PES2UG24CS042-pes-vcs$ ./pes add file1.txt file2.txt
akash@LAPTOP-NAL20LH5:/mnt/d/OS_orange/PES2UG24CS042-pes-vcs$ ./pes status
Staged changes:
  staged:   file1.txt
  staged:   file2.txt

Unstaged changes:
(nothing to show)

Untracked files:
untracked: .gitignore
untracked: commit.c
untracked: commit.h
untracked: file.txt
untracked: fix.py
untracked: fix2.py
untracked: fix_index_add.py
untracked: index.c
untracked: index.h
untracked: Makefile
untracked: object.c
untracked: pes.c
untracked: pes.h
untracked: README.md
untracked: solve.py
untracked: solve2.py
untracked: test_objects.c
untracked: test_sequence.sh
untracked: test_tree.c
untracked: tree.c
untracked: tree.h
```

3B

```
akash@LAPTOP-NAL20LH5:/mnt/d/OS_orange/PES2UG24CS042-pes-vcs$ cat .pes/index
304444 2cf8d83d9ee29543b34a87727421fdec7e3f3a183d337639025de576db9ebb4 1776343867 6 file1.txt
304444 e00c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f 1776343874 6 file2.txt
```

4A

```
akash@LAPTOP-NAL20LH5:~/PES2UG24CS042-pes-vcs$ ./pes init
Initialized empty PES repository in .pes/
akash@LAPTOP-NAL20LH5:~/PES2UG24CS042-pes-vcs$ echo "Hello" > hello.txt
akash@LAPTOP-NAL20LH5:~/PES2UG24CS042-pes-vcs$ ./pes add hello.txt
akash@LAPTOP-NAL20LH5:~/PES2UG24CS042-pes-vcs$ ./pes commit -m "Initial commit"
Committed: de544b46ae03... Initial commit
akash@LAPTOP-NAL20LH5:~/PES2UG24CS042-pes-vcs$ echo "World" >> hello.txt
akash@LAPTOP-NAL20LH5:~/PES2UG24CS042-pes-vcs$ ./pes add hello.txt
akash@LAPTOP-NAL20LH5:~/PES2UG24CS042-pes-vcs$ ./pes commit -m "Add world"
Committed: 43f6ab318517... Add world
akash@LAPTOP-NAL20LH5:~/PES2UG24CS042-pes-vcs$ echo "Goodbye" > bye.txt
akash@LAPTOP-NAL20LH5:~/PES2UG24CS042-pes-vcs$ ./pes add bye.txt
akash@LAPTOP-NAL20LH5:~/PES2UG24CS042-pes-vcs$ ./pes commit -m "Add farewell"
Committed: 3b0b7f00ea03... Add farewell
akash@LAPTOP-NAL20LH5:~/PES2UG24CS042-pes-vcs$ ./pes log
commit 3b0b7f00ea031f9ea20f18e156f884f51fbb8d5c8bb87d11852062e942c352e4
Author: Akash Singh <PES2UG24CS042>
Date: 1776352730

    Add farewell

commit 43f6ab318517aceb54a7d49e060acb6f21d27452695f1871665668e2d3c63e28
Author: Akash Singh <PES2UG24CS042>
Date: 1776352715

    Add world

commit de544b46ae03e8ec65f2b33c9e4152d7c430f1da5579fb8cf9a826eb5fde9092
Author: Akash Singh <PES2UG24CS042>
Date: 1776352692

    Initial commit farewell

commit 9f9826d29ce0a2da8c995b06d8975b43d0aa7eb88b0f28fe33bb1e61fde3e609
Author: Akash Singh <PES2UG24CS042>
Date: 1776352566

    final2566

finalS0Q
akash@LAPTOP-NAL20LH5:~/PES2UG24CS042-pes-vcs$ |
```

4B

```
akash@LAPTOP-NAL20LH5:~/PES2UG24CS042-pes-vcs$ find .pes -type f | sort
.pes/HEAD
.pes/index
.pes/objects/05/c8d777d0be85748b6788d1b5caf7d69754e6e68060c265ca21cbd6c05e5f43
.pes/objects/2c/f8d83d9ee29543b34a87727421fdec7e3f3a183d337639025de576db9ebb4
.pes/objects/3b/0b7f00ea031f9ea20f18e156f884f51fbb8d5c8bb87d11852062e942c352e4
.pes/objects/43/f6ab318517aceb54a7d49e060acb6f21d27452695f1871665668e2d3c63e28
.pes/objects/5b/ce0b2732efe7ab4dff59fb271f4b0ca1e3ed766db23b0c7b611f52502ffc3c
.pes/objects/66/224663d23e6f4d9de9e2c7e6d8764305a92a3830a1a52d3d5f4aa8007b5c39
.pes/objects/68/27d53d89f5f84060b293129bdc91427adb58be1acd82e8a5c428ea7304f1b4
.pes/objects/9f/9826d29ce0a2da8c995b06d8975b43d0aa7eb88b0f28fe33bb1e61fde3e609
.pes/objects/ad/aebb678eed9cdff00e74d36a144cb377d4575b742e3e7e127aad68a24f9bae7
.pes/objects/be/fc883dfad4d0e70ea21cd0adca03856138a39d324502e34936b85b3922c27c
.pes/objects/de/544b46ae03e8ec65f2b33c9e4152d7c430f1da5579fb8cf9a826eb5fde9092
.pes/objects/e6/7ed66bcb4a708250a89f315d4d8bb92703343c84df834438880e13a68652bc
.pes/refs/heads/main
akash@LAPTOP-NAL20LH5:~/PES2UG24CS042-pes-vcs$ |
```

4C

```
akash@LAPTOP-NAL20LH5:~/PES2UG24CS042-pes-vcs$ cat .pes/refs/heads/main
3b0b7f00ea031f9ea20f18e156f884f51fbb8d5c8bb87d11852062e942c352e4
akash@LAPTOP-NAL20LH5:~/PES2UG24CS042-pes-vcs$ cat .pes/HEAD
ref: refs/heads/main
akash@LAPTOP-NAL20LH5:~/PES2UG24CS042-pes-vcs$ |
```

Phase 5 and 6

5.1

Update HEAD to point to .pes/refs/heads/<branch>

Read commit hash from branch file

Load commit -> get tree -> restore all files in working directory

5.2

For each tracked file:

Get hash from index

Compute hash of working file

If hashes differ -> file is modified

If modified file also differs in target branch -> refuse checkout

5.3

HEAD points directly to a commit

New commits are created but **not linked to any branch**

They may become unreachable

Recovery:

Create a branch using commit hash

6.1

Start from all branch heads

Traverse commits -> trees -> blobs

Mark reachable objects (use **HashSet**)

Delete unvisited objects Estimate:

Need to visit hundreds of thousands of objects

6.2

During commit, new objects may not yet be referenced

GC may delete them -> commit becomes invalid Git

avoids this by:

using locks delaying deletion
(grace period) ensuring atomic
updates